

A Cubic with No Critical Element over $C(S^1)$

Counterexample to two C^* -algebraic mean value conjectures

Counterexample packet for arXiv:2206.08154

26 August 2026

Status: candidate full counterexample, likely valid. There is a degree-three polynomial over the commutative C^* -algebra $C(S^1)$ whose derivative has no zero in the algebra. At an explicitly chosen noncritical point, this contradicts the existential conclusion common to the C^* -algebraic Smale and Dubinin–Sugawa conjectures, including their strong forms.

1 Source and exact conjectures

The source is K. Mahesh Krishna, *C^* -algebraic Smale Mean Value Conjecture and Dubinin–Sugawa Dual Mean Value Conjecture*, arXiv:2206.08154 (2022) [1]. For a commutative C^* -algebra $\mathcal{A}$ and $P(x) = \prod_{j=1}^n (x - a_j)$, it defines a critical element to be $w \in \mathcal{A}$ with $P'(w) = 0$.

Conjecture 2.1 says that if z is not critical, there is a critical $w \in \mathcal{A}$ satisfying the proposed Smale upper estimate. Conjecture 3.1 says that, under the same hypotheses, there is a critical $w \in \mathcal{A}$ satisfying the proposed Dubinin–Sugawa lower estimate. Conjectures 2.2 and 3.2 make stronger C^* -order assertions but retain the same existence claim. The formal statements are on source pages 4–5 and 7; both complete conjectures also appear in the abstract on source page 1.

**C*-ALGEBRAIC SMALE MEAN VALUE CONJECTURE AND DUBININ-SUGAWA
DUAL MEAN VALUE CONJECTURE**

K. MAHESH KRISHNA

Post Doctoral Fellow

Statistics and Mathematics Unit

Indian Statistical Institute, Bangalore Centre

Karnataka 560 059 India

Email: kmaheshak@gmail.com

Date: June 17, 2022

Abstract: Based on Smale mean value conjecture [*Bull. Amer. Math. Soc.*, 1981] and Dubinin-Sugawa dual mean value conjecture [*Proc. Japan Acad. Ser. A Math. Sci.*, 2009] we formulate the following conjectures.

C*-algebraic Smale Mean Value Conjecture : Let $\mathcal{A}$ be a commutative C*-algebra. Let $P(z) = (z - a_1) \cdots (z - a_n)$ be a polynomial of degree $n \geq 2$ over $\mathcal{A}$, $a_1, \dots, a_n \in \mathcal{A}$. If $z \in \mathcal{A}$ is not a critical point of P , then there exists a critical point $w \in \mathcal{A}$ of P such that

$$\frac{\|P(z) - P(w)\|}{\|z - w\|} \leq 1 \|P'(z)\|$$

or

$$\frac{\|P(z) - P(w)\|}{\|z - w\|} \leq \frac{n-1}{n} \|P'(z)\| = \frac{\deg(P) - 1}{\deg(P)} \|P'(z)\|.$$

C*-algebraic Dubinin-Sugawa Dual Mean Value Conjecture : Let $\mathcal{A}$ be a commutative C*-algebra. Let $P(z) = (z - a_1) \cdots (z - a_n)$ be a polynomial of degree $n \geq 2$ over $\mathcal{A}$, $a_1, \dots, a_n \in \mathcal{A}$. If $z \in \mathcal{A}$ is not a critical point of P , then there exists a critical point $w \in \mathcal{A}$ of P such that

$$\frac{\|P'(z)\|}{\deg(P)} = \frac{\|P'(z)\|}{n} \leq \frac{\|P(z) - P(w)\|}{\|z - w\|}.$$

We show that (even a strong form of) C*-algebraic Smale mean value conjecture and C*-algebraic Dubinin-Sugawa dual mean value conjecture hold for degree 2 C*-algebraic polynomials over commutative C*-algebras.

Keywords: C*-algebra, Smale mean value theorem, Smale mean value conjecture, Dubinin-Sugawa dual mean value theorem, Dubinin-Sugawa dual mean value conjecture.

Mathematics Subject Classification (2020): 46L05, 30C10.

Figure 1: The source abstract on PDF page 1, containing the two conjectures in full and noting that the paper proves only degree two.

2 Proof intuition

For a cubic, solving $P'(w) = 0$ requires choosing a square root of its discriminant. Over the scalar field this is automatic. Over $C(S^1)$, an element is a continuous function, so a root of the derivative must amount to a continuous choice of one of the two scalar critical points in every fiber.

We arrange for the derivative discriminant to trace a nonvanishing loop of odd winding number. It has two distinct square roots in every individual fiber but no continuous square root around the circle: traversing the circle exchanges the two roots. Hence the polynomial has no critical element at all. Choosing $z = 0$ with $P'(0)$ invertible meets the conjectures' hypothesis, while their required w cannot exist; the numerical inequalities never enter.

3 The construction and proof

Write $\mathbb{T} = \{\zeta \in \mathbb{C} : |\zeta| = 1\}$ and let $u \in C(\mathbb{T})$ be the coordinate function $u(\zeta) = \zeta$. Set

$$\rho = e^{i\pi/3} = \frac{1 + i\sqrt{3}}{2}, \quad \varepsilon = \frac{1}{4}, \quad c = \rho + \varepsilon u.$$

Lemma 1 (Odd winding has no continuous square root). *If $q \in C(\mathbb{T})$ is nowhere zero and $\text{wind}(q)$ is odd, then there is no $h \in C(\mathbb{T})$ with $h^2 = q$.*

Proof. Such an h would also be nowhere zero. Winding number is additive under pointwise multiplication, so $\text{wind}(q) = \text{wind}(h^2) = 2 \text{wind}(h)$, an even integer. $\square$

Theorem 1. *In the unital commutative C^* -algebra $\mathcal{A} = C(\mathbb{T})$, the cubic*

$$P(x) = x(x - 1)(x - c)$$

has no critical element. Nevertheless $z = 0$ is not critical. Consequently Conjectures 2.1, 2.2, 3.1, and 3.2 of [1] are false in degree three.

Proof. Expanding in the commutative algebra gives

$$P(x) = x^3 - (1 + c)x^2 + cx, \quad P'(x) = 3x^2 - 2(1 + c)x + c.$$

At $z = 0$ we have $P'(0) = c$. Pointwise,

$$|c(\zeta)| = |\rho + \varepsilon\zeta| \geq 1 - \varepsilon = \frac{3}{4},$$

so c is invertible in $C(\mathbb{T})$; in particular, z is not critical.

Suppose that $w \in C(\mathbb{T})$ were a critical element. From $P'(w) = 0$ we get

$$9w^2 = 6(1 + c)w - 3c.$$

Thus, with $h = 3w - (1 + c)$,

$$\begin{aligned} h^2 &= 9w^2 - 6(1 + c)w + (1 + c)^2 \\ &= (1 + c)^2 - 3c = c^2 - c + 1. \end{aligned}$$

Because $\rho^2 - \rho + 1 = 0$ and $2\rho - 1 = i\sqrt{3}$,

$$q := c^2 - c + 1 = \varepsilon u(i\sqrt{3} + \varepsilon u). \tag{1}$$

The second factor never vanishes since $|i\sqrt{3} + \varepsilon\zeta| \geq \sqrt{3} - \varepsilon > 0$. Its image is a circle centered at $i\sqrt{3}$ of radius ε , which does not surround the origin, so its winding number is zero. The factor εu has winding number one. Therefore

$$\text{wind}(q) = 1 + 0 = 1.$$

Equation (1) and Lemma 1 contradict $h^2 = q$. Hence P' has no zero in $C(\mathbb{T})$.

The polynomial has degree three and the required factor form with roots $0, 1, c \in \mathcal{A}$. Each of the four cited conjectures applies at a noncritical z and concludes that some critical $w \in \mathcal{A}$ exists. Our admissible $z = 0$ has no such w , so all four conclusions fail. $\square$

Fiberwise interpretation. The obstruction is genuinely global, not scalar. At every $\zeta \in \mathbb{T}$, the scalar derivative has the two distinct roots

$$w_{\pm}(\zeta) = \frac{1 + c(\zeta) \pm \sqrt{q(\zeta)}}{3},$$

because $q(\zeta) \neq 0$. Odd monodromy prevents either choice from defining a continuous function on the whole circle.

4 Verification, novelty, and scope

Verification. The argument is exact and uses only polynomial algebra and the elementary winding-number obstruction. The included checker samples 16,384 points, verifies (1), the lower bounds for $|c|$ and $|q|$, winding number one, and the scalar derivative roots. It is not used as proof.

Bounded novelty search. The four run indexes were searched for arXiv:2206.08154, the exact conjecture titles, $C(S^1)$, critical points, continuous square roots, monodromy, and winding. Exact-title, exact-phrase, author/id, counterexample, degree-three, and close-variant arXiv/web searches through 26 August 2026 found the source paper but no later paper claiming this construction or an exact resolution. Novelty is plausible, not certified.

Scope and review. This is a full counterexample to the unrestricted commutative C^* -algebra statements in Conjectures 2.1 and 3.1 and, a fortiori, to their proposed strong forms. It does not address the classical scalar Smale or Dubinin–Sugawa conjectures. It also leaves open versions over algebras whose spectrum has no square-root monodromy obstruction. Human review should check the derivative identity, the source’s critical-point definition, and the winding calculation.

References

- [1] K. Mahesh Krishna, *C^* -algebraic Smale Mean Value Conjecture and Dubinin–Sugawa Dual Mean Value Conjecture*, [arXiv:2206.08154](https://arxiv.org/abs/2206.08154), 2022.